MASTER'S COURSE

Petrophysics

LECTURE 2

How do forces and deformation become stress and strain?

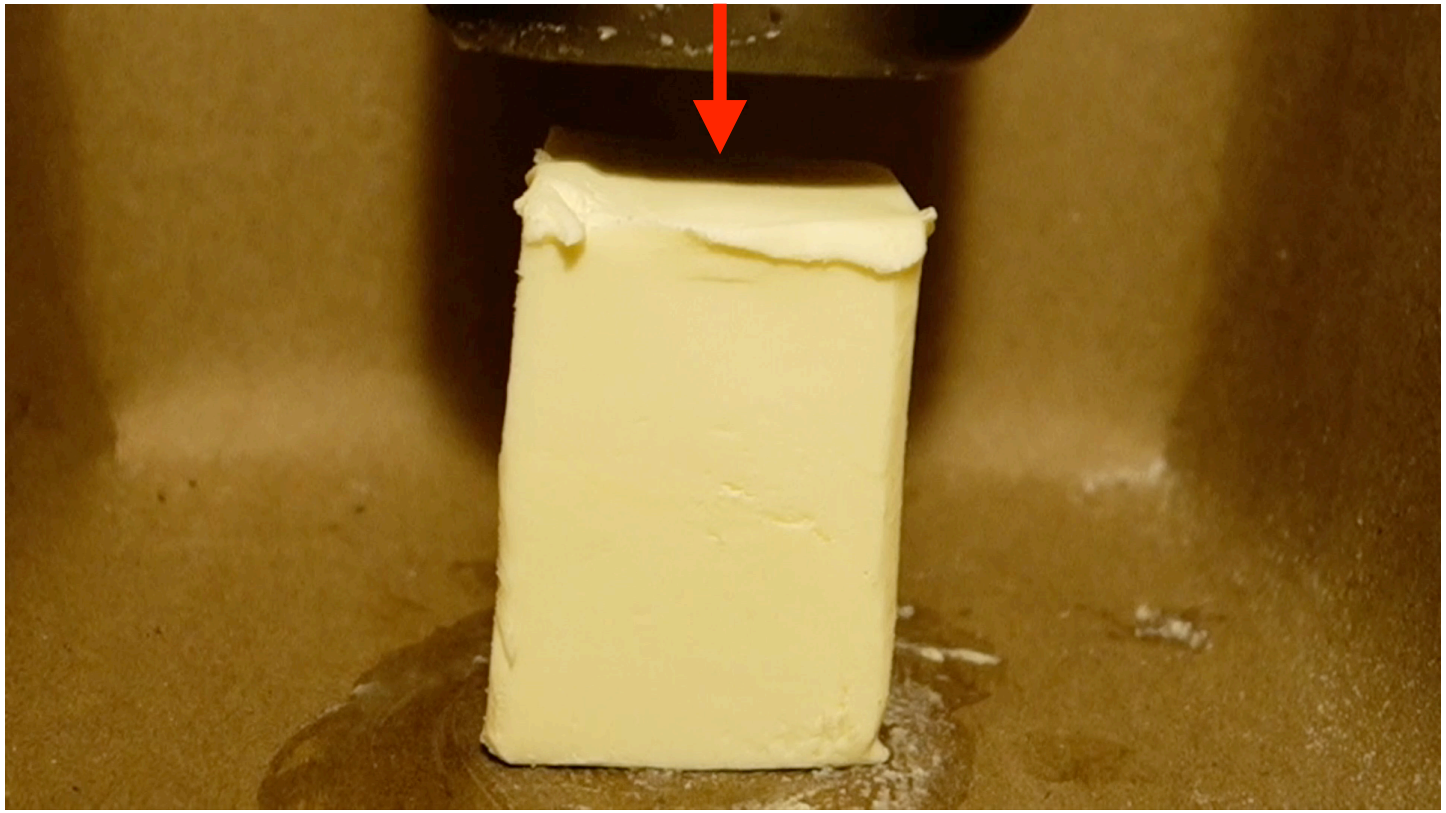


Marco Scuderi · Sapienza Università di Roma

OPENING PROBLEM

Why does vertical loading produce an inclined fracture?

Force



Predict the deformation, then describe only what you observe.

OPENING PROBLEM

Why does vertical loading produce an inclined fracture?



BEFORE



AFTER

Predict the deformation, then describe only what you observe.

OPENING PROBLEM

Why does vertical loading produce an inclined fracture?



BEFORE



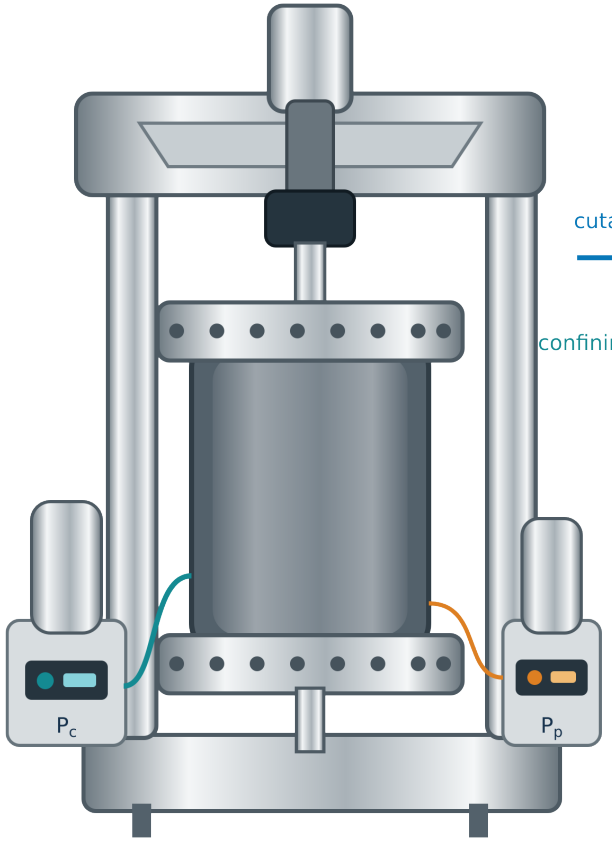
AFTER

Before explaining failure, we must define what the experiment actually **measures**.

THINK · PAIR · SHARE

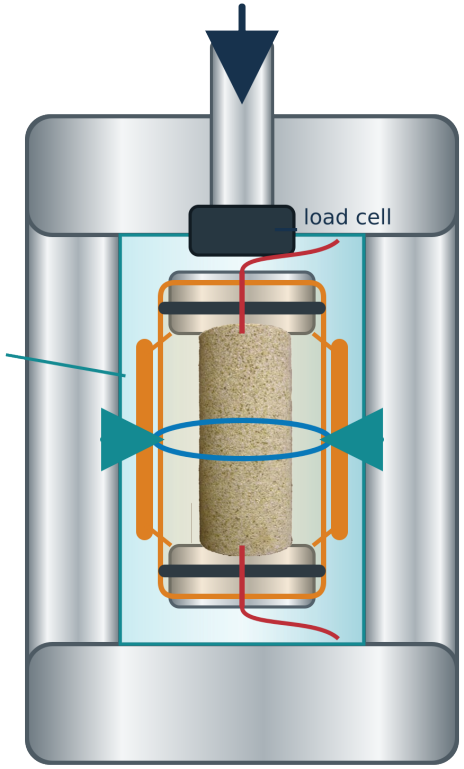
Before calculating, decide what we measure

HIGH-PRESSURE TRIAXIAL APPARATUS



cutaway

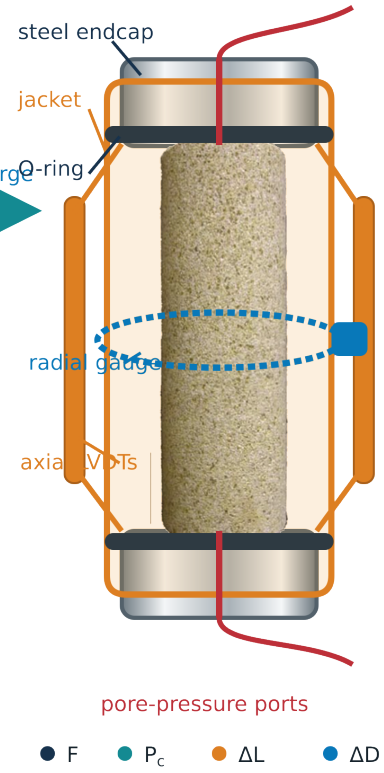
PRESSURE VESSEL CUTAWAY



P_c acts on the jacket, not directly on the rock

enlarge

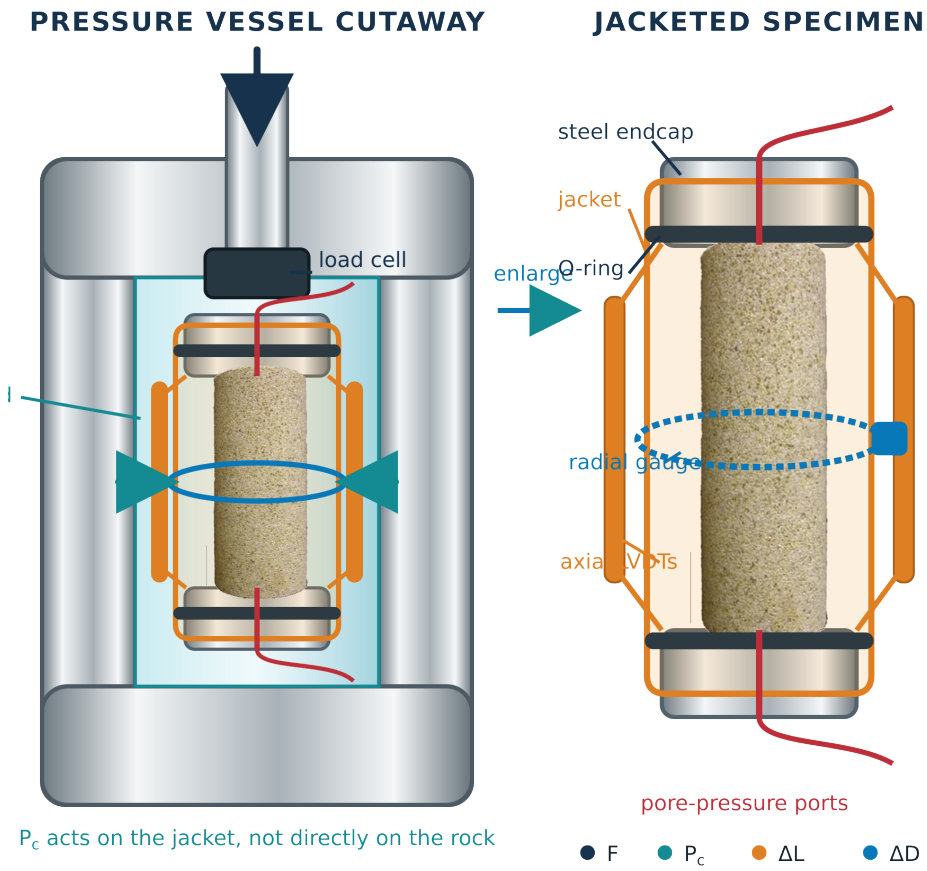
JACKETED SPECIMEN



Which quantities do we measure directly? Which do we calculate?

GEOMETRY	DEFORMATION
Diameter and length [mm]	Axial and radial change [mm]
FORCE	**BOUNDARY**
Load cell [N]	Confining pressure [MPa]

Before calculating, decide what we measure



P_c acts on the jacket, not directly on the rock

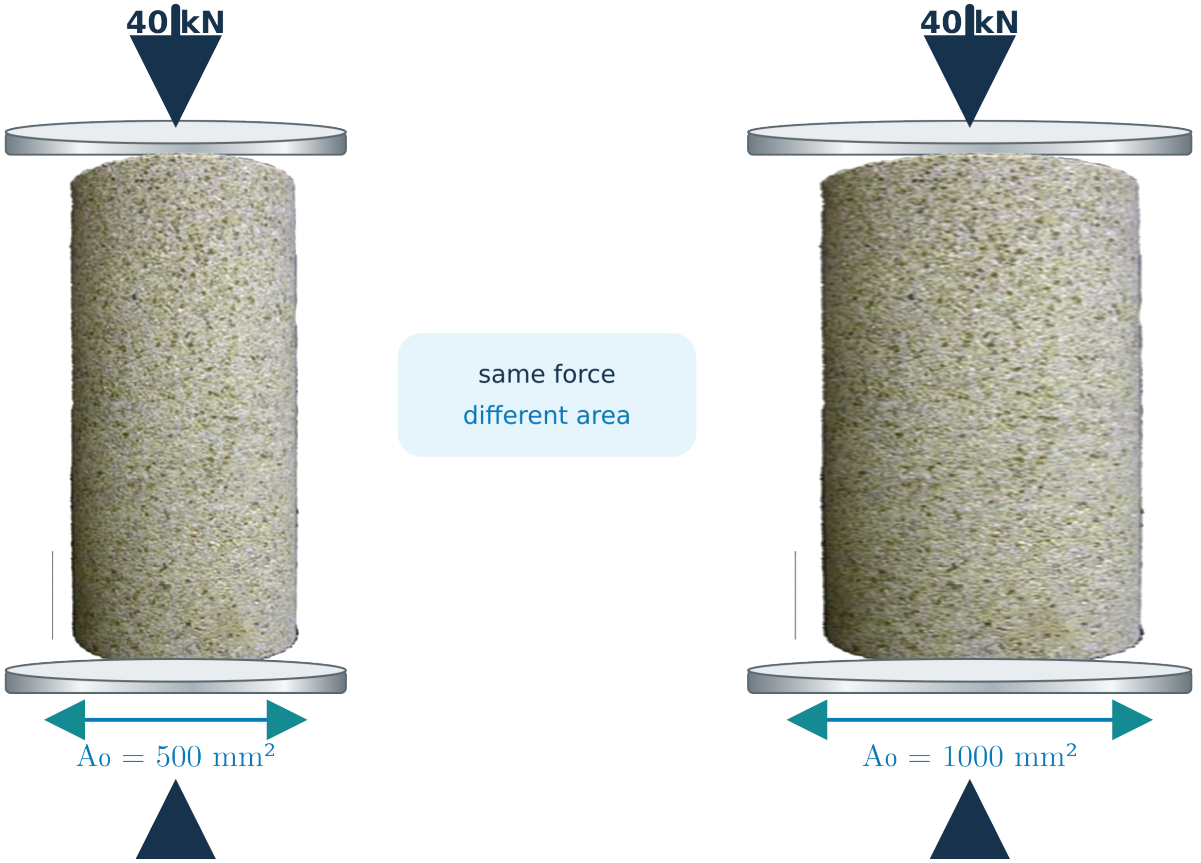
Which quantities do we measure directly? Which do we calculate?

Measured directly	Calculated
Force [N]	Stress (σ) [MPa]
Displacement ΔL [mm]	Strain ε
Diameter D_0	Cross-sectional area A_0
Confining pressure Pc	Principal and mean stresses

Which of these quantities depends on the specimen geometry?

Does equal force means equal stress ?

Activity 1



Which specimen carries the larger average axial stress?

Stress normalizes force by area



Nominal axial stress

$$\sigma_a = \frac{F}{A_0}$$

$$A_0 = \frac{\pi D^2}{4}$$

Use the original area for nominal stress and state that choice.

1 N/mm² = 1 MPa